

Harrison Crosse

📍 Arlington, VA, US ✉ harrison@crosse.dev 📞 703-472-7202 in hcrosse 🌐 hcrosse

EMPLOYMENT

Docker

Senior Software Engineer 2025 - Present

- Co-designed Marlin, Docker's event ingestion and processing platform, for tens of thousands of events per second. Established its initial SDK and ingestion architecture and guided expansion into managed stream processing and audit-log analytics.
- Built Marlin client SDKs for browser, Node.js, and Rust using shared protobuf data contracts for consistent event schemas. Productionized the SDKs with batching, retries, integration tests, and release automation.
- Designed Marlin's privacy architecture and implemented annotation-driven PII handling across client SDKs, ingestion workers, and Iceberg storage, supporting Docker's ISO 27701 certification.
- Hardened Marlin's production ingestion with schema preflight validation, version-gated migrations, DLQs, and production observability.

Calendly

Senior Data Engineer 2024 - 2025
Data Engineer 2022 - 2024

- Built Calendly's new data platform from the ground up, migrating Postgres and event data from Redshift to BigQuery and establishing Airflow, CI/CD, permissions, data quality, and developer tooling.
- Processed 50 TB of malformed Segment replay data with Spark to complete its migration to BigQuery.
- Built and operated production Flink pipelines processing thousands of records per second with hundreds of gigabytes of state. Productionized them with observability, blue-green deployments, data-quality gates, and E2E tests.
- Led the rollout of Segment Data Contracts across hundreds of event types, building schema generation, CLI tooling, and CI/CD to keep application and data-platform schemas synchronized.

Amobee

Software Engineer, Data Systems 2021 - 2022

- Migrated clickstream processing from batch Spark jobs to Spark Structured Streaming, making event data continuously available to downstream consumers.
- Moved Spark workloads from EMR to Kubernetes with Spark Operator, standardizing deployments and eliminating EMR service charges.

Sayari Labs

Data Engineer 2020 - 2021

- Built web-scraping and fastText NER pipelines to transform corporate registry data into structured entities and social graphs.
- Migrated Sayari's cloud infrastructure from GCP to AWS GovCloud using Terraform and Docker.

EDUCATION

University of Mary Washington

B.S. Computer Science 2019
Minor in Data Science

SELECTED WORK

DataFusion / Ballista

Upstreamed non-blocking distributed shuffle execution by exposing async batch partitioning in DataFusion and offloading Ballista shuffle writes from query workers.

DataFusion #21341 Ballista #1537

Apache Iceberg / Arrow Go

Improved Parquet writer compatibility by adding table-level page-version configuration to Iceberg and fixing invalid root-schema serialization in Arrow Go.

Iceberg #15700 Arrow Go #723

TOOLS

LANGUAGES

Bash C Go Java Python
Rust SQL TypeScript

DATA SYSTEMS

Airflow Arrow BigQuery
ClickHouse dbt Debezium
Flink Iceberg Kafka
Parquet PostgreSQL Protobuf
Snowflake Spark

INFRASTRUCTURE

Argo AWS Docker GCP
GitHub Actions Grafana Helm
Kubernetes Linux Terraform

Booz Allen Hamilton

Data Scientist

2019 - 2020

- Modeled Marine Corps unit performance for the Inspector General using clustering and gradient-boosted trees.